

Basic Statistical Concepts

Moments · Estimators · MoM & MLE · Inference

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Learning objectives

By the end of this chapter, you will be able to:

- interpret the *moments* of a distribution (mean, variance, skewness, kurtosis);
- compute classical *point estimators* (sample mean, variance, ACF) and discuss their properties (bias, consistency);
- apply two estimation methods: *Method of Moments* and *Maximum Likelihood*;
- describe the *sampling distribution* of standard estimators under normality;
- run a basic *statistical inference* procedure (confidence interval, hypothesis test, p -value).

Chapter outline

1. Moments of a distribution
2. Classical estimators and sampling distributions
3. Estimation methods: MoM and MLE
4. Multivariate data
5. Statistical inference (CI, hypothesis testing)

Moments of a distribution

1. Raw and central moments

Definition – Moments

Let X be a random variable with density f and mean $\mu = \mathbb{E}[X]$.

- **Raw ℓ -th moment:** $m'_\ell = \mathbb{E}[X^\ell] = \int x^\ell f(x) dx$.
- **Central ℓ -th moment:** $m_\ell = \mathbb{E}[(X - \mu)^\ell]$.

Standard quantities:

$$\mu = m'_1, \quad \sigma^2 = m_2 = \text{Var}[X], \quad S(X) = \frac{m_3}{\sigma^3}, \quad K(X) = \frac{m_4}{\sigma^4}.$$

1. Numerical example: $X \sim \mathcal{N}(2, 4)$

Theoretical moments:

$$\mu = 2, \quad \sigma^2 = 4, \quad S(X) = 0, \quad K(X) = 3.$$

Empirical estimates from $n = 10\,000$ Monte-Carlo draws (seed 42):

$$\hat{\mu} \approx 2.005, \quad \hat{\sigma}^2 \approx 3.987, \quad \hat{S} \approx -0.013, \quad \hat{K} \approx 2.996.$$

Skewness measures asymmetry; *kurtosis* measures tail-heaviness. A normal distribution has $S = 0$ and $K = 3$ (excess kurtosis $K - 3 = 0$).

Classical estimators

2. Sample mean and sample variance

Given i.i.d. data $X_1, \dots, X_n$:

$$\bar{X}_n = \frac{1}{n} \sum_{i=1}^n X_i, \quad S_n^2 = \frac{1}{n-1} \sum_{i=1}^n (X_i - \bar{X}_n)^2.$$

Proposition – Properties

Under i.i.d. assumption with finite second moments:

- **Unbiased:** $\mathbb{E}[\bar{X}_n] = \mu$ and $\mathbb{E}[S_n^2] = \sigma^2$.
- **Consistent:** $\bar{X}_n \xrightarrow{p} \mu$ and $S_n^2 \xrightarrow{p} \sigma^2$ (Law of Large Numbers).

2. Sampling distribution under normality

If $X_1, \dots, X_n \stackrel{\text{i.i.d.}}{\sim} \mathcal{N}(\mu, \sigma^2)$:

Theorem

$$\bar{X}_n \sim \mathcal{N}\left(\mu, \frac{\sigma^2}{n}\right), \quad \frac{(n-1) S_n^2}{\sigma^2} \sim \chi_{n-1}^2, \quad \frac{\bar{X}_n - \mu}{S_n/\sqrt{n}} \sim t_{n-1}.$$

- $\bar{X}_n$ and S_n^2 are *independent* under normality (Cochran's theorem).
- These distributions are the basis of z-tests, t -tests and χ^2 -tests.

2. Central Limit Theorem

Theorem – CLT

For i.i.d. $X_1, \dots, X_n$ with $\mathbb{E}[X_i] = \mu$ and $\text{Var}[X_i] = \sigma^2 < \infty$:

$$\frac{\bar{X}_n - \mu}{\sigma/\sqrt{n}} \xrightarrow{d} \mathcal{N}(0, 1).$$

Even when X_i is *not* normal, the sample mean is approximately normal for large n .

Rule of thumb: $n \geq 30$.

Estimation methods

3. Method of Moments (MoM)

Definition – MoM

To estimate $\theta \in \mathbb{R}^k$, solve the system

$$\mathbb{E}_{\theta}[X^p] = \frac{1}{n} \sum_{i=1}^n X_i^p, \quad p = 1, \dots, k.$$

Example: $\mathcal{N}(\mu, \sigma^2)$.

$$\mu = \mathbb{E}[X] \Rightarrow \hat{\mu}_{\text{MoM}} = \bar{X}_n,$$

$$\sigma^2 = \mathbb{E}[X^2] - \mu^2 \Rightarrow \hat{\sigma}_{\text{MoM}}^2 = \frac{1}{n} \sum (X_i - \bar{X}_n)^2.$$

Note: this differs from S_n^2 by a factor $n/(n-1)$.

3. Maximum Likelihood (MLE)

Definition – Likelihood and MLE

Likelihood: $L(\theta) = \prod_{i=1}^n f(x_i; \theta)$ for i.i.d. data. Log-likelihood: $\ell(\theta) = \log L(\theta)$. The MLE is

$$\hat{\theta}_{\text{MLE}} = \arg \max_{\theta} \ell(\theta).$$

Example: $\mathcal{N}(\mu, \sigma^2)$. Solving $\partial \ell / \partial \mu = 0$ and $\partial \ell / \partial \sigma^2 = 0$:

$$\hat{\mu}_{\text{MLE}} = \bar{X}_n, \quad \hat{\sigma}_{\text{MLE}}^2 = \frac{1}{n} \sum (X_i - \bar{X}_n)^2.$$

Same as MoM for the Gaussian. Both are biased for σ^2 .

3. Asymptotic normality of the MLE

Theorem

Under regularity conditions (identifiability, smoothness, finite Fisher information $I(\theta_0)$):

$$\sqrt{n}(\hat{\theta}_n - \theta_0) \xrightarrow{d} \mathcal{N}(0, I(\theta_0)^{-1}).$$

- MLE is **asymptotically efficient** (reaches the Cramér–Rao bound).
- MoM is simpler but generally less efficient.
- Both are consistent.

Multivariate data

4. Joint, marginal, conditional

Joint density: $f(x, y)$.

$$f_X(x) = \int f(x, y) dy, \quad f_{Y|X}(y | x) = \frac{f(x, y)}{f_X(x)},$$

$$f(x, y) = f_{Y|X}(y | x) f_X(x).$$

Independence: $f(x, y) = f_X(x) f_Y(y)$.

$$\mathbb{E}[XY] = \mathbb{E}[X]\mathbb{E}[Y], \quad \text{Cov}(X, Y) = 0.$$

4. Sample covariance and correlation

Given paired data $(X_i, Y_i)_{i=1}^n$:

$$\widehat{\text{Cov}}(X, Y) = \frac{1}{n-1} \sum_{i=1}^n (X_i - \bar{X})(Y_i - \bar{Y}),$$

$$\widehat{\text{Corr}}(X, Y) = \frac{\widehat{\text{Cov}}(X, Y)}{\hat{\sigma}_X \hat{\sigma}_Y} \in [-1, 1].$$

Correlation captures *linear* dependence only. Two variables can have $\text{Corr}(X, Y) = 0$ and still be strongly dependent (e.g. $Y = X^2$ with X symmetric).

Statistical inference

5. Confidence interval for the mean

Setting: $X_1, \dots, X_n \stackrel{\text{i.i.d.}}{\sim} \mathcal{N}(\mu, \sigma^2)$. σ **known** (z-interval):

$$\text{CI}_{1-\alpha}(\mu) = \left[\bar{X}_n - z_{1-\alpha/2} \frac{\sigma}{\sqrt{n}}, \bar{X}_n + z_{1-\alpha/2} \frac{\sigma}{\sqrt{n}} \right].$$

σ **unknown** (t-interval):

$$\text{CI}_{1-\alpha}(\mu) = \left[\bar{X}_n - t_{n-1, 1-\alpha/2} \frac{S_n}{\sqrt{n}}, \bar{X}_n + t_{n-1, 1-\alpha/2} \frac{S_n}{\sqrt{n}} \right].$$

Reading: probability $1 - \alpha$ that the true μ falls in the interval (over repeated samples).

5. Hypothesis testing

Definition – Two-sided test on the mean

$H_0 : \mu = \mu_0$ vs. $H_1 : \mu \neq \mu_0$. Test statistic $T = (\bar{X}_n - \mu_0)/(S_n/\sqrt{n}) \sim t_{n-1}$ under H_0 . Reject H_0 at level α if $|T| > t_{n-1, 1-\alpha/2}$.

- **p-value:** probability under H_0 of observing something at least as extreme as the data.
- **Type I error** (α): rejecting H_0 when true.
- **Type II error** (β): not rejecting H_0 when false.
- **Power:** $1 - \beta$ — sensitivity to detect H_1 .

5. Numerical example

We sample $n = 50$ Atlanta monthly temperatures (in $^{\circ}\text{F}$). Sample statistics: $\bar{X}_{50} = 61.2$, $S_{50} = 16.8$. Test $H_0 : \mu = 60$ vs. $H_1 : \mu \neq 60$ at $\alpha = 0.05$.

$$T = \frac{61.2 - 60}{16.8/\sqrt{50}} \approx 0.505, \quad t_{49,0.975} \approx 2.010.$$

Since $|T| = 0.505 < 2.010$, **do not reject** H_0 . p -value ≈ 0.616 .

The 95% CI for μ is $[61.2 - 4.78, 61.2 + 4.78] = [56.4, 66.0]$, which contains 60 — consistent with the test conclusion.

Wrap-up

Key takeaways

- The *moments* characterise a distribution (mean, variance, skewness, kurtosis).
- *Classical estimators* $(\bar{X}, S^2)$ are unbiased and consistent under i.i.d. assumptions.
- *MoM* and *MLE* coincide for the Gaussian; in general MLE is asymptotically efficient, MoM is simpler.
- Under normality, $\bar{X} \sim \mathcal{N}(\mu, \sigma^2/n)$ and $(n-1)S^2/\sigma^2 \sim \chi_{n-1}^2$.
- *CLT* extends asymptotic normality to non-Gaussian data.
- Confidence intervals and hypothesis tests are built on the sampling distribution of the estimator.

Chapter 2. Time-series *decomposition*: trend and seasonality.

- Classical additive and multiplicative decomposition.
- Trend estimation by moving averages and parametric models.
- Seasonality estimation.
- Residual analysis.

Thank you.

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